

QIUZHEN QUALIFY EXAM FOR ARTIFICIAL INTELLIGENCE

MOCK EXAMINATION NO. 4

Instructions

- This examination consists of four sections, each worth 33 points. Choose exactly three sections and answer all questions in the selected sections.
 - Write your name and student ID on the answer sheet for 1 additional point. The maximum possible score is 100.
 - State the three selected sections at the beginning of the answer sheet. If more than three sections are attempted, only the first three will be graded.
 - Unless stated otherwise, justify every claim and show the essential derivation. Unsupported final formulas receive limited credit.
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A. Machine Learning Theory

- [8 pts.]** Determine whether each statement is true or false. Give a proof or a concrete counterexample; a bare answer receives no credit.
 - [2 pts.]** If a classifier has zero training error on an i.i.d. sample, its population error converges to zero as the sample size grows, regardless of how the classifier was selected.
 - [2 pts.]** Every symmetric function $k(x, z)$ is a valid positive-semidefinite kernel.
 - [2 pts.]** For separable data, unregularized logistic regression always has a finite MLE.
 - [2 pts.]** If k_1 and k_2 are positive-semidefinite kernels, then $k_1 + k_2$ and $k_1 k_2$ are positive-semidefinite kernels.
- [7 pts.]** Let the instance space contain $2m$ distinct points. A deterministic learning algorithm receives the labels of a uniformly selected subset of m points and returns a binary classifier. The target labeling is drawn uniformly from all 2^{2m} binary labelings. Prove that the expected error of the returned classifier on the m unseen points is $1/2$. Explain what this construction does and does not imply about learning with structural assumptions.
- [8 pts.]** A researcher has M classifiers whose predictions are independent fair coin flips, independent of all labels. The same balanced validation set of n examples is used to select the classifier with the smallest validation error; ties are broken arbitrarily.
 - [3 pts.]** Compute the probability that at least one classifier has zero validation error.
 - [2 pts.]** Determine the selected classifier's expected error on a fresh test example and explain the discrepancy when M is large.
 - [3 pts.]** Give an evaluation protocol that yields an unbiased final performance estimate, and state a uniform high-probability correction for selecting among a fixed set of M candidates.

4. [10 pts.] Let $\mathcal{H}$ be an RKHS with kernel k , and consider

$$\min_{f \in \mathcal{H}} \Psi(f(x_1), \dots, f(x_m)) + \Omega(\|f\|_{\mathcal{H}}),$$

where a minimizer exists and $\Omega : [0, \infty) \rightarrow \mathbb{R}$ is strictly increasing. Prove that every minimizer lies in $\text{span}\{k(x_i, \cdot) : 1 \leq i \leq m\}$. Derive the resulting finite-dimensional optimization problem and its kernel Gram matrix. State what changes if Ω is only nondecreasing.

B. Deep Learning and Reinforcement Learning

1. [8 pts.] Consider a width- n feedforward network with independent weights $W_{ij}^{(\ell)}$ of mean zero and variance σ_w^2/n , zero biases, and activations $h^{(\ell)} = \phi(W^{(\ell)}h^{(\ell-1)})$. Assume coordinates are approximately independent and identically distributed at initialization.

- (a) [3 pts.] Derive the forward second-moment recursion.
- (b) [3 pts.] Derive the corresponding backward gradient-variance multiplier and specialize both recursions to ReLU under a symmetric preactivation distribution.
- (c) [2 pts.] Explain why making all hidden units in a layer have identical incoming and outgoing parameters, including all-zero initialization, creates a symmetry that deterministic gradient descent cannot break.

2. [7 pts.] An overparameterized autoencoder is trained on a finite dataset using only reconstruction error and has enough capacity to implement the identity map.

- (a) [3 pts.] Show why perfect reconstruction alone does not imply that the latent representation captures useful structure, and construct a degenerate solution.
- (b) [2 pts.] Formulate a denoising-autoencoder objective and derive the Bayes-optimal reconstruction function under squared loss.
- (c) [2 pts.] Explain how a bottleneck or corruption changes the learning problem, including one limitation of each remedy.

3. [8 pts.] For one-step MAML, a task loss is split into support and query losses. The inner update and meta-objective are

$$\theta'_\tau = \theta - \alpha \nabla L_\tau^{\text{sup}}(\theta), \quad J(\theta) = \mathbb{E}_\tau L_\tau^{\text{qry}}(\theta'_\tau).$$

Derive the exact meta-gradient, including the Hessian term, and the first-order MAML approximation. Then construct a one-dimensional pair of quadratic support/query losses for which the one-step support update increases the query loss. State the assumption about task relatedness that the counterexample violates.

4. [10 pts.] For a finite discounted MDP with $0 \leq \gamma < 1$:

- (a) [3 pts.] Write the Bellman expectation operator T^π and Bellman optimality operator T^* on action-value functions, and prove that each is a γ -contraction in the sup norm.
- (b) [3 pts.] Explain why the fixed point of T^π evaluates a fixed policy whereas the fixed point of T^* solves optimal control. Give a two-action counterexample to replacing the policy expectation by a maximum during policy evaluation.
- (c) [4 pts.] State exact policy iteration and prove that its policy-improvement step cannot decrease the value function. Include the tie case.

C. Optimization Methods in Artificial Intelligence

1. [9 pts.] Determine whether each statement is true or false and prove the claim or give a counterexample.

- (a) [3 pts.] Every stationary point of a differentiable convex function is a global minimizer.
- (b) [3 pts.] The gradient of every differentiable L -smooth function is a monotone operator.
- (c) [3 pts.] For any proper closed convex g , the proximal map prox_g is firmly nonexpansive.

2. [8 pts.] Consider the scalar objective $f(x) = x^2/2$ and Adam with $\beta_1 = \beta_2 = 0$, $\epsilon = 0$, and a fixed step size $\alpha > 0$:

$$m_t = g_t, \quad v_t = g_t^2, \quad x_{t+1} = x_t - \alpha \frac{m_t}{\sqrt{v_t}}, \quad g_t = \nabla f(x_t).$$

Starting from $x_0 = \alpha/2$, derive the entire iterate sequence and show that the method does not converge to the minimizer. Identify which common convergence safeguard is absent and explain the role of $\epsilon > 0$ near a stationary point.

3. [8 pts.] Let $f(x) = \frac{1}{2}x^\top Qx - b^\top x$ with $Q \succ 0$, and let $P \succ 0$ be a fixed preconditioner. Consider

$$x^{k+1} = x^k - \gamma P \nabla f(x^k).$$

Transform the iteration into symmetric coordinates, derive the exact spectral condition on γ for convergence, and give the optimal constant step size and worst-case contraction factor in terms of the extreme eigenvalues of $P^{1/2}QP^{1/2}$. Explain when preconditioning improves the rate.

4. [8 pts.] Let $X \subset \mathbb{R}^d$ be convex with ℓ_∞ diameter at most D_∞ . Online convex losses f_t have subgradients g_t with $\|g_t\|_2 \leq G$. Diagonal AdaGrad uses

$$H_t = \text{diag} \left(\epsilon + \sqrt{\sum_{s=1}^t g_{s,1}^2}, \dots, \epsilon + \sqrt{\sum_{s=1}^t g_{s,d}^2} \right)$$

and $x_{t+1} = \Pi_X^{H_t}(x_t - \eta H_t^{-1} g_t)$. Derive a regret bound that adapts to the coordinate-wise accumulated gradients, and compare it with the $O(DG\sqrt{T})$ Euclidean bound when gradients are sparse.

D. Natural Language Processing

1. [9 pts.] A linear-chain CRF assigns

$$p_\theta(y | x) = \frac{\exp(\theta^\top F(x, y))}{Z_\theta(x)}, \quad Z_\theta(x) = \sum_{y'} \exp(\theta^\top F(x, y')).$$

For one labeled sequence, derive the negative log-likelihood gradient and Hessian. Prove convexity in θ , characterize when strict convexity fails, and state how the required feature expectations are computed for a first-order chain.

2. [8 pts.] Let self-attention without positional encodings map $X \in \mathbb{R}^{L \times d}$ to

$$A(X) = \text{softmax}_{\text{row}} \left(\frac{XW_QW_K^\top X^\top}{\sqrt{d_k}} \right) XW_V.$$

Prove permutation equivariance. Then show by an explicit length-two permutation that a fixed causal mask destroys equivariance to arbitrary token permutations. State which permutations, if any, remain compatible with causal order.

3. [7 pts.] For a preference pair (x, y^+, y^-) , define

$$z_\theta = \beta \left[\log \frac{\pi_\theta(y^+ | x)}{\pi_{\text{ref}}(y^+ | x)} - \log \frac{\pi_\theta(y^- | x)}{\pi_{\text{ref}}(y^- | x)} \right], \quad \ell(\theta) = -\log \sigma(z_\theta).$$

Derive $\nabla_\theta \ell$. Prove that multiplying both reference probabilities for this pair by the same positive constant leaves the loss unchanged. Explain why this pairwise invariance does not make the reference policy irrelevant.

4. [9 pts.] An entity linker must jointly resolve two mentions m_1, m_2 . Candidate variables E_1, E_2 have unary scores $u_i(e)$ from a text encoder. The knowledge graph supplies a weighted compatibility feature $r(e_1, e_2)$. Define a two-variable conditional log-linear/MLN model, derive its partition function, marginals, and MAP rule, and give exact computational complexity in the candidate-set sizes. Explain how the pairwise factor can correct locally preferred but globally inconsistent links, and analyze one failure caused by a missing knowledge-graph edge.